

Executive Summary

Apex Precision Components (APC) faces a critical choice: retrench into a constrained legacy infrastructure or modernize to capture the \$8M projected revenue from SmartLogistics Solutions (SLS). With storage at 70% capacity, hardware averaging 4.5 years old, and a rigid July 2026 deadline, the current path is unsustainable.

To meet contractual obligations without disrupting the \$12,000/hour production line, I recommend **Option 3: Cloud Migration Strategy**. This approach solves immediate capacity limits, aligns with SLS's cloud-native architecture, and protects the manufacturing edge. It shifts IT resources from low-value maintenance to strategic integration.

Analysis: AWS Cloud Adoption Framework (CAF) Perspectives

A holistic transformation is required to bridge the gap between APC's legacy stability and SLS's innovation across six perspectives:

- **Business Perspective:** Our goal to "predict demand patterns" is currently blocked by infrastructure limits. Migrating non-manufacturing workloads unlocks the scalability needed to deliver the "integrated supply chain analytics" product on time. We must bridge the **Innovation Management** gap, shifting focus from maintenance to strategic data monetization.
- **People Perspective:** Merging SLS’s cloud-native team with APC’s legacy IT requires **Culture Evolution** to prevent operational friction. We must address the **Cloud Fluency** gap by upskilling staff to support high-value integration rather than routine hardware mechanics.
- **Governance Perspective:** We must shift from a risky \$800k capital outlay to a variable OpEx model (**Cloud Financial Management**), aligning costs with revenue. Establishing **Risk Management** frameworks for data residency will satisfy automotive client requirements.
- **Platform Perspective:** With rack space full, on-premises expansion is a dead end. Cloud migration resolves this via **Provisioning & Orchestration**. To protect manufacturing, we will adopt an **Edge Computing** strategy: keeping real-time MES local for millisecond latency while piping data to the cloud for analysis.
- **Security Perspective:** Cloud **Defense in Depth** (VPCs) offers superior isolation for ITAR data compared to our current physical perimeter, which hinders remote work. A centralized **Identity and Access Management (IAM)** system will resolve fragmentation and secure the hybrid workforce.
- **Operations Perspective:** Replace the costly Tennessee DR site with **Availability & Continuity** via multi-Region cloud architecture. Unified **Observability** will provide a "single pane of glass" to monitor the hybrid ecosystem.

Strategic Recommendation & Comparative Analysis

I strongly recommend **Option 3: Cloud Migration Strategy**.

Option	Viability	Strategic Justification
Option 1: On-Prem	None	Strategic Misalignment. An 18-month buildout guarantees missing the July 2026 deadline. It forces a cloud-native acquisition backward, reducing the agility we acquired.
Option 2: Hybrid	Low	Operational Risk. Managing two distinct stacks doubles the security surface area and creates a permanent technical and cultural silo, hindering integration.

Option	Viability	Strategic Justification
Option 3: Cloud	High	Strategic Alignment. Solves the rack/CapEx crisis instantly. It aligns with the acquisition's tech stack, modernizes legacy IT, and meets the critical timeline.

Justification: Option 3 mitigates the CEO's concerns regarding platform reliance not by avoidance, but by **portability** (using Kubernetes containers) and **edge security**. It ensures the factory floor remains independent of internet connectivity while leveraging the cloud for enterprise scale.

Transformation Journey (CAF Roadmap)

To execute this strategy while managing risk, we will follow the CAF phases:

- 1. **Envision:** Align leadership on cloud value and business outcomes (Immediate).
- 2. **Align:** Form a Cloud Center of Excellence (CCoE) to bridge talent and assess gaps (Month 1).
- 3. **Launch:** Pilot non-critical workloads (Email, Backups) to build confidence (Months 2-3).
- 4. **Scale:** Migrate ERP and integrate SLS analytics to hit the July 2026 target (Months 4-9).